



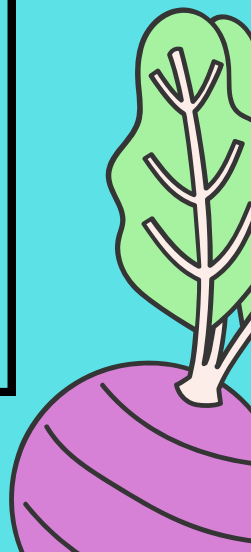
DIETARY & NUTRITION PLANNER

Course: Front End Development Frameworks & UI Engineering (FEDF&UI)

Team Members

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PROJECT OVERVIEW

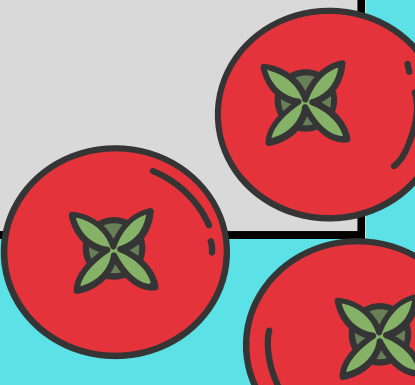
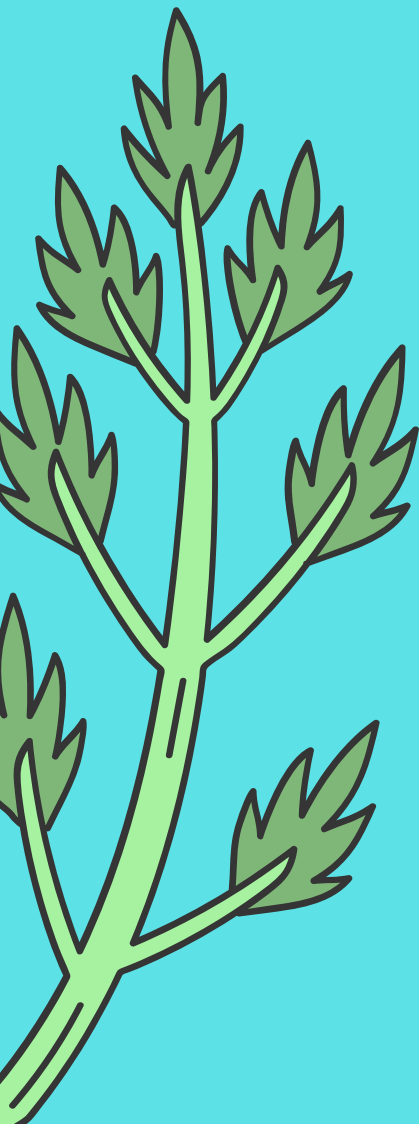
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The Dietary & Nutrition Planner is a web-based healthcare application that helps hospitals efficiently manage patient nutrition, meal planning, and kitchen operations. It provides personalized diet plans based on patient health conditions while improving communication between doctors, dietitians, and kitchen staff.

2

Modules

- Patient Management
- Diet Planning
- Meal Scheduling
- Kitchen Management
- Reports & Analytics
- Notifications
- AI Nutrition Assistant



PROBLEM STATEMENT

Problems

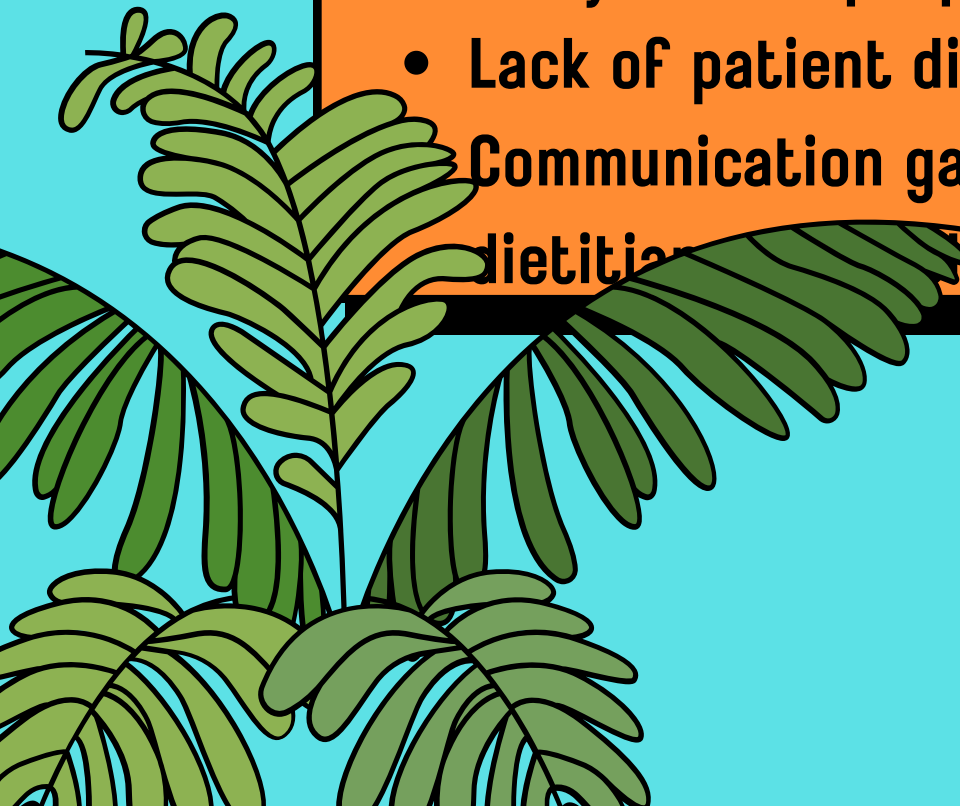
Hospitals face several nutrition management challenges:

- Manual diet planning
- Incorrect calorie calculations
- Delayed meal preparation
- Lack of patient diet tracking

Communication gap between
dietitians and chefs

Proposed Solution

Develop a React-based Dietary & Nutrition Planner that automates meal planning, calorie calculation, kitchen order management, and nutrition monitoring.



OBJECTIVES & FEATURES

Objectives

- Automate meal planning
- Generate personalized diet charts
- Improve patient nutrition management
- Reduce manual workload
- Increase hospital efficiency

Features

- Secure Login
- Patient Records
- Diet Recommendation
- Meal Scheduling
- Kitchen Dashboard
- Nutrition Reports
- AI Meal Suggestions

SYSTEM REQUIREMENTS

Software	Purpose
VS Code	Development
React.js	Frontend
TypeScript	Programming
Vite	Build Tool
Node.js	Runtime
GitHub	Version Control

TECH STACK

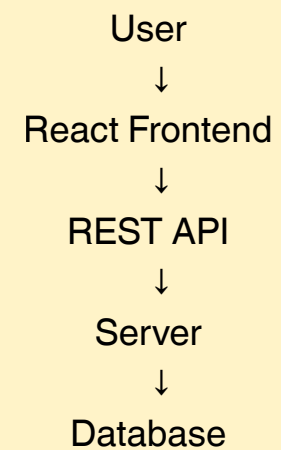
FRONTEND

- React.js
- TypeScript
- Tailwind CSS
- Lucide React Icons

BACKEND / API

- Node.js
- REST APIs
- Fetch API
- Gemini AI Integration

Architecture



APPLICATION LAYOUT

Login

Dashboard

Patient Management

Diet Planner

Meal Scheduler

Kitchen Orders

Reports

AI Nutrition Assistant

Notifications

NutriPlanPro

OPERATIONS CONSOLE

Comprehensive Dashboard

Patient Profiles & Calculators

AI Nutrient Planner

Clinical Kitchen Queue

Reports & Diagnostics Log

DS

Dr. Sarah Peterson

Doctor (Cardiology)

CARE COMPLIANCE MODE

Hide Guide

Patient Dietary Console Dashboard

Monitoring inpatients allergy logs and medical menu synchronisation instantly.

Interactive NutriPlan Console Flow

Master the entire patient clinical nutrition loop in 4 simplified stage trials.

TRIAL GUIDE: DietitianDoctorKitchen ChefAdmin

01 / STAGE

Admit Inpatients

Register age, medical conditions & allergens. Daily calorie target BMR formula runs automatically.

02 / STAGE

Generate AI Diet

Select patient, run clinical meal builder. Gemini scans allergies to formulate clean recipes.

03 / STAGE

Physician Sign-off

Doctor credentials verify the food formulation so preparation and kitchen logs can be launched.

04 / STAGE

Kitchen Prep Queue

Approved meals sync automatically. Cooks prepare, finalize, and mark meals as hand-delivered.

🕒

Dietitian Mode:

Go to [Patient Profiles](#) to add an inpatient, then navigate to [AI Nutrient Planner](#) to draft safety-inspected therapeutic diets using artificial intelligence.

ACTIVE INPATIENTS

4

PENDING APPROVALS

1

ACTIVE COOK ORDERS

2

PREVENTED ALLERGIES

7

Active Patient Diet Tracking

View all patient profiles >

PATIENT	WARD / BED	ALLERGY RISK	CALORIE TARGET	SYNC STATE
James Vance P-101 (Male, 64y)	Cardiology Room A-302	Shellfish, Soy	1800 kcal	🟢 Synchronized
Emma Lin	Obstetrics & Gynecology			

Actionable Care Alerts

Hospital notification logs of dietary preventions and chef updates.

New Patient Admitted

02:30 pm

Emma Lin (Ward: OBGYN, Room 108) admitted with Gestational Diabetes. Allergy warning: Peanuts, Gluten.

REACT COMPONENT FOLDER STRUCTURE

```
src
|
|— components
|   |— Login.tsx
|   |— Dashboard.tsx
|   |— Patients.tsx
|   |— DietPlanner.tsx
|   |— MealScheduler.tsx
|   |— Kitchen.tsx
|   |— Reports.tsx
|   └─ Notifications.tsx
|
|— App.tsx
|— main.tsx
└─ types.ts
```


ADDITIONAL CODE SNIPPETS

```
export type UserRole = 'Dietitian' | 'Doctor' | 'KitchenStaff' | 'Admin' | 'Patient';

export interface User {
  id: string;
  name: string;
  role: UserRole;
  email: string;
  avatarUrl?: string;
  department?: string;
}
```

```
export interface Patient {
  id: string;
  name: string;
  age: number;
  gender: 'Male' | 'Female';
  weight: number; // in kg
  height: number; // in cm
  activityLevel: 'Sedentary' | 'Light' | 'Moderate' | 'Active';
  medicalConditions: string[]; // e.g. Diabetes, Hypertension, Celiac Disease, renal
  allergies: string[]; // e.g. Peanuts, Dairy, Gluten, Egg, Shellfish
  roomNumber: string;
  ward: string;
  admissionDate: string;
  targetCalories: number; // Calculated daily caloric need
  macronutrients: {
    protein: number; // in grams
    carbs: number; // in grams
    fat: number; // in grams
  };
}
```

APP DEMO & SCREENSHOTS

OPERATIONS CONSOLE

Comprehensive Dashboard

Patient Profiles & Calculators

AI Nutrient Planner

Clinical Kitchen Queue

Reports & Diagnostics Log

Search patients by name / ID...

All Wards

+ Add Admitted Patient

P-101

James Vance

Room A-302 • Ward Cardiology

Gender / Age
Male, 64 yrs

Weight / Height
82 kg, 175 cm

MEDICAL CONDITIONS

Hypertension Type 2 Diabetes

STRICT ALLERGIES

Shellfish Soy

21%

Daily Calorie Target
1800 kcal
Served: 580 kcal

PRO CARB FAT
90g 200g 55g

P-102

Emma Lin

Room B-108 • Ward Obstetrics & Gynecology

Gender / Age
Female, 29 yrs

Weight / Height
68 kg, 162 cm

MEDICAL CONDITIONS

Gestational Diabetes Post-Op Recovery

STRICT ALLERGIES

Peanuts Gluten

0%

Daily Calorie Target
2100 kcal

PRO CARB FAT
110g 220g 65g

P-103

Marcus Brody

Room A-105 • Ward General Medicine

Gender / Age
Male, 45 yrs

Weight / Height
90 kg, 182 cm

MEDICAL CONDITIONS

Hypothyroidism Hypercholesterolemia

STRICT ALLERGIES

Dairy Gluten

0%

Daily Calorie Target
1950 kcal

PRO CARB FAT
95g 220g 50g

P-104

Sophia Martinez

Room C-214 • Ward Geriatrics

Gender / Age
Female, 72 yrs

Weight / Height
54 kg, 155 cm

MEDICAL CONDITIONS

CKD Stage 3 Mild Osteoporosis

STRICT ALLERGIES

Walnuts

DS

Dr. Sarah Peterson
Doctor (Cardiology)

Automated Meal Planning

Leverage Gemini to craft sterile nutritional dishes conforming to the client's medical guidelines.

Select Hospital Patient

-- Choose admitted patient --

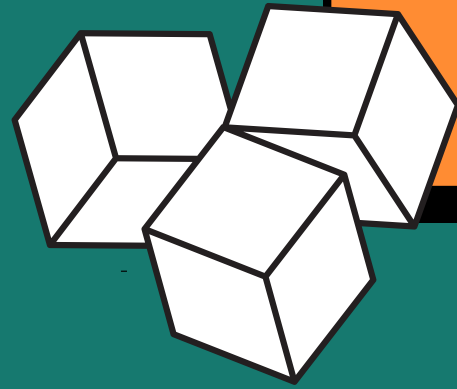
Please select an admitted patient to begin.

Active Planning Workspace

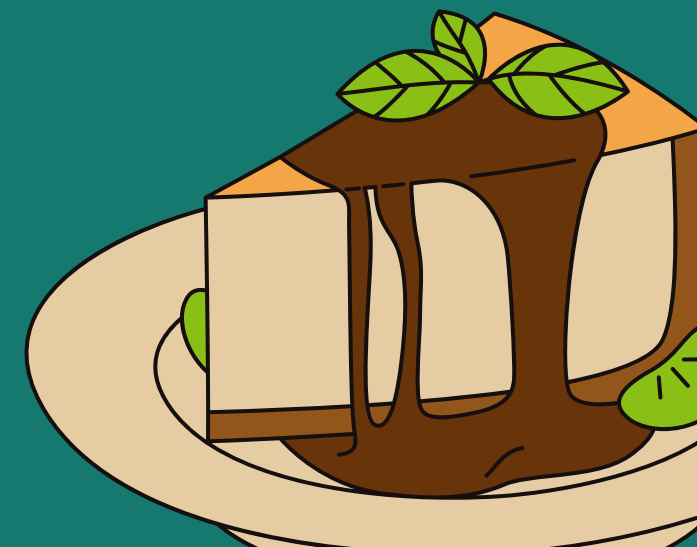
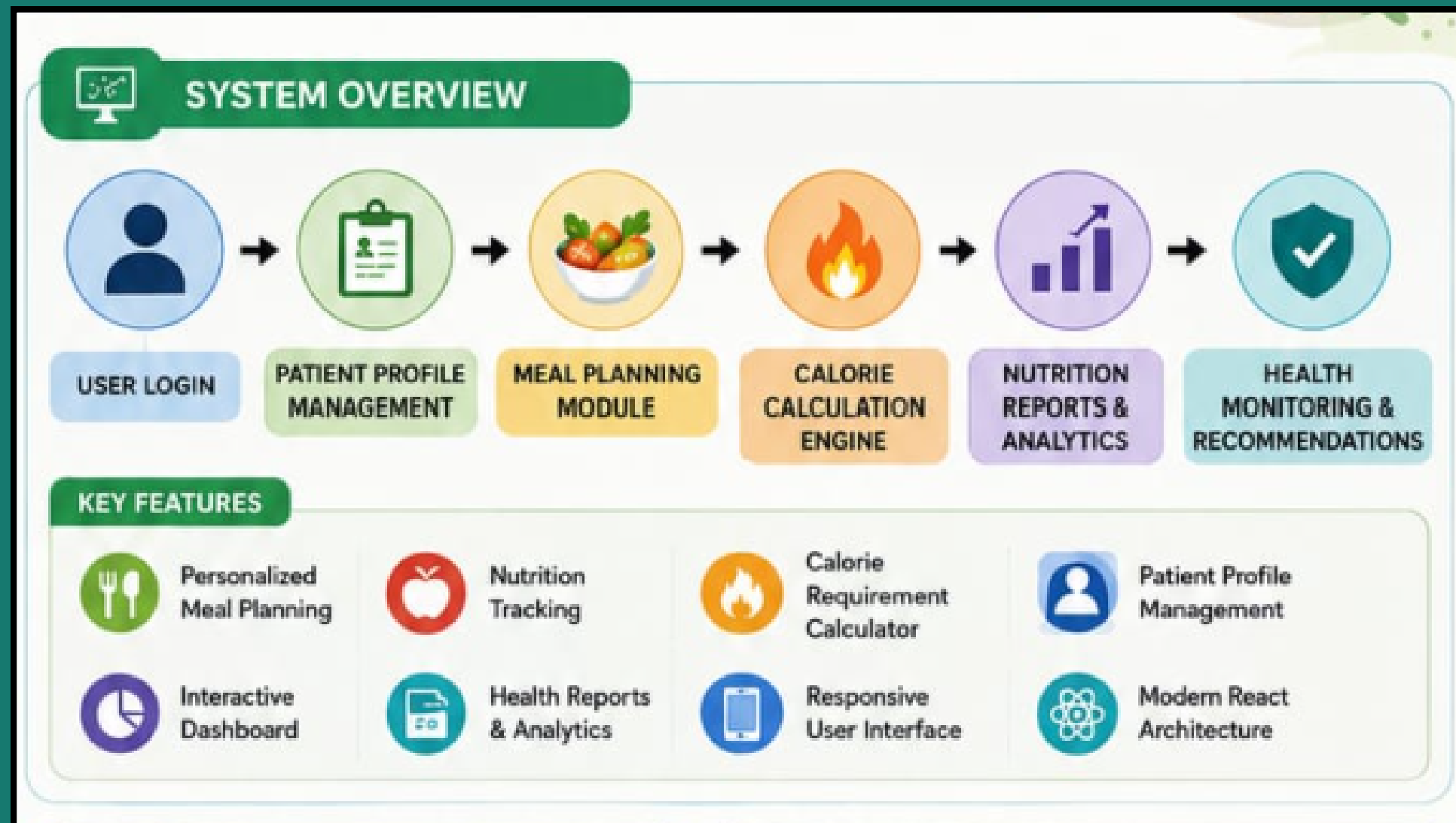
Select an admitted patient on the left panel to execute automatic safety assessments and formulate calorie budgets.

Clinical Meal Plans & Approval States

PLAN ID	PATIENT NAME	TOTAL CALORIES	NUTRIENTS (P/C/F)	ALLERGY CHECK	WORKFLOW STATE	ACTIONS / VERIFICATION
MP-201	James Vance	1600 kcal	97g / 155g / 57g	✓ Clean	✓ Approved	Signed by Dr. Sarah Peterson
MP-202	Marcus Brody	1860 kcal	98g / 163g / 75g	✓ Clean	Pending Doctor Signature	<button>Approve / Sign</button> <button>Reject</button>



WORKING WORKFLOW



Conclusion

Project Summary

- Developed a hospital dietary management platform.
- Automated patient meal planning.
- Improved nutrition tracking.
- Simplified kitchen workflow.
- Reduced manual effort.

Key Technical Learnings

- React Components
- TypeScript
- React Hooks
- REST APIs
- State Management
- Responsive UI Design
- AI Integration

